

Urban mobility analysis: Optimizing the flow of public traffic through Big data processing

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Abstract—Public transit systems in Santiago de Chile, comprising the RED bus network and Metro, generate continuous, high-volume streams of smart-card validation data that traditional Relational Database Managements Systems are ill-equipped to process. This work proposes a Big Data architecture for urban mobility analysis that combines historical DTPM validation records with static GTFS spatial data to infer Origin-Destination matrices under a tap-in-only data model, addressing the absence of explicit destination information through Alighting Estimation techniques. Beyond this retrospective reconstruction, the proposed pipeline integrates live GTFS-Realtime and GPS fleet feeds to characterize the system’s current operational state, enabling the detection of congestion bottlenecks as they emerge rather than days after the fact.

By articulating a Batch Processing layer for large-scale historical inference alongside a Stream Processing layer for near-real-time monitoring, this study aims to shift transport management from static, monthly reporting toward proactive and predictive operational decision-making, ultimately supporting reduce operational cost for service providers and shorter travel times for users.

1. Introduction

Modern public transit system such as RED and Metro in Santiago generate a continuous massive stream of transactional data. Every passenger interaction validating a card at a turnstile produces a record characterized by high density. These smart card validation systems act as the primary sensor network for urban mobility, capturing millions of daily events that reflect the dynamic movement of people across the metropolitan area.

Despite the wealth of data generated by urban transit, transport authorities face major operational and computational challenges. First, long travel times and suboptimal route configurations severely erode the quality of life for daily citizens, understanding these extensive displacements patterns is critical for system improvement. Also high passenger congestion dynamically collapses specific stations and stops, specially during peak hours, the lack of rapid, scalable data analysis prevents real-

time reallocation to alleviate these critical bottlenecks. Furthermore traditional Relational Database Management System are computationally ill-equipped to process and query gigabytes of high-density data efficiently [1], processing millions of daily transaction requires transitioning toward distributed Big Data architectures (such as Batch and Stream Processing ecosystems) capable of handling volume and high generation velocity[1]

Transitioning from traditional analytical approaches to Big Data ecosystems allows transport engineering to shift from reactive decisions-making-based on static monthly reports to proactive and predictive management [1]

By processing massive historical validation records alongside spatial transit models, it becomes possible to accurately infer Origin-Destination matrices and optimize vehicle operational frequencies. This optimization reduces operational expenditures for service providers while directly minimizing daily travel times for citizens, returning valuable time to commuters and delivering a clear social impact.

2. Datasets

2.1. Smart Card Validation Records (DTPM)

The primary transactional source will be the historical validation records provided by "Directorio de Transporte Público Metropolitano" (DTPM) [1]. This dataset contains massive, anonymized records of passenger transactions upon entering the system (RED).

- **Structure and key variables:** The information is presented in heavy tabular formats and includes the exact timestamp of the transaction, an anonymized card identifier, and the ID of the boarding stop or station.
- **Data limitation and technical justification:** The system operates under a tap-in only logic, meaning the passenger’s actual destination is a nonexistent data point in the raw database. To accurately map congestion, it is necessary to reconstruct trips by applying Alighting Estimation algorithms. This requires processing the total daily volume by grouping

records by card ID, sorting them chronologically, and calculating the destination by inferring spatial proximity to the user’s next trip. This massive operation of sorting, crossing, and sequential inference over millions of rows overwhelms traditional relational databases, making the use of Batch Processing architectures strictly necessary.

2.2. General Transit Feed Specification (GTFS) Data

To provide spatial context to the transactional data and enable the destination inference algorithms, information from the GTFS standard—obtained from the DTPM and the State’s Open Data Portal [2] will be integrated.

- **Structure and key variables:** It consists of inter-related text files that provide the exact geographic coordinates (latitude and longitude) of each stop and station, the spatial shape of the routes, and the scheduled transit times.
- **Relevance:** When a user makes their last trip of the day, the algorithm does not have a “next trip” to deduce their destination. In these cases, reconstructing the flow requires crossing the card’s temporal history with the physical route of the bus modeled in the GTFS. By merging the high-volume validation records with the spatial density of the GTFS through multidimensional join operations, it becomes possible to geographically map the passenger flow and locate hidden bottlenecks within the network.

2.3. Congestion GroundTruth and Validation Metrics

A recurring limitation in Big Data pipelines for urban mobility is the absence of an explicit, verifiable definition of *congestion*. Without one, bottleneck is no independent basis to confirm whether those anomalies correspond to real operational stress experienced by passengers. This subsection defines congestion operationally and identifies reference datasets against which the pipeline’s output can be validated.

Operational definition. For the purposes of this study, congestion at a given stop, corridor or time window is modeled as a function of two computable dimensions:

$$C = f(\text{Occupancy}, \text{WaitTime})$$

Occupancy is approximated through the density of tap-in validations per vehicle per unit time, derived directly from the DTPM smart-card records described in Section 2.1. Wait time is approximated through headway, the interval between successive arrivals of a given service at a stop, obtained by contrasting scheduled headway from the static GTFS feed (Section 2.2) against observed headway from

the GTFS-Realtime stream.

External validation sources. Because both components of C are derived from the same data ecosystem used to build the model, they are insufficient on their own to demonstrate that the detected congestion reflects reality rather than an artifact of the inference pipeline. Two independent DTPM-published sources are proposed to address this:

- **Occupancy Rate Measurement Methodology:** The DTPM maintains a standardized field-measurement protocol for bus occupancy rates at fixed points and stops, using structured forms completed by trained personnel. Where historical records exist for a subset of corridors and dates, they provide an authority-recognized ground truth against which the occupancy proxy from validation density can be statistically contrasted.
- **User Satisfaction and Service Evaluation Studies:** periodically the DTPM publishes user evaluation studies covering satisfaction with operators and the system overall, which capture subjective perceptions of crowding and waiting time by corridor.

Validation criterion. A corridor-period is considered a *validated* congestion event when the pipeline’s composite score C exceeds a defined threshold and at least one external source (field-measured occupancy or user-reported dissatisfaction) concurs for the same corridor and agreement rate between data-driven detection and independently observed operational conditions, rather than presenting bottleneck detection as an unverified byproduct of the Big Data architecture.

References

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